

Multi-source urban signals

311 | Weather
POI | Land use

311

Weather

POI

Land use

Leakage-safe panel

NTA x Week x Category

History at t

-> Event at t+1

shifted history only

LightGBM + XGBoost

2015-22 2022-23 2024-25

Train Validation
Future test

Category calibrated alerts

Validation-only
category thresholds
score -> threshold -> alert

Risk alerts

SHAP review